

Conformal Selection of Counterfactual Explanations: From Minimal to Robust Recourse

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Abstract

Counterfactual explanations are commonly generated by identifying minimally perturbed instances that change a model prediction. However, minimal perturbations do not necessarily correspond to realistic, representative, or robust forms of recourse. In particular, counterfactuals located close to the decision boundary may satisfy the desired prediction outcome while still exhibiting atypical feature configurations or relying on unusual model reasoning patterns. This paper proposes a conformal framework for selecting counterfactual explanations based on their conformity to the target-class distribution. Rather than selecting the closest generated counterfactual, the proposed approach selects the candidate with highest conformal p-value according to conformal anomaly detection. Conformity is evaluated both in feature space and in SHAP space, enabling robustness to be assessed not only in terms of geometric similarity, but also in terms of explanatory consistency. Experiments on twelve benchmark datasets demonstrate that substantially more conforming counterfactual explanations can often be obtained with only modest increases in perturbation magnitude. In particular, the results show that counterfactuals with similar feature-space proximity may nevertheless differ substantially in SHAP-space conformity, suggesting that minimally perturbed counterfactuals may rely on atypical explanatory patterns. Qualitative examples further illustrate how conformity-based selection can produce explanations that appear more representative of genuine target-class instances. The proposed framework further highlights how conformity-based reasoning can extend beyond predictive uncertainty estimation and provide principled tools for assessing the quality and consistency of AI explanations.

Keywords: Counterfactuals, Conformal anomaly detection, XAI, SHAP

1. Introduction.

Counterfactual explanations have emerged as a prominent tool for interpreting machine learning models by providing actionable insights in the form of “what-if” scenarios. Given an instance and its predicted label, a counterfactual explanation specifies the minimal changes required to alter the model’s decision. For example, in a credit scoring setting, a counterfactual might indicate that increasing income by a certain amount would lead to loan approval. This notion of *minimal recourse*, i.e., the smallest perturbation needed to flip the prediction, has become the standard paradigm for counterfactual explanations.

While minimal recourse is intuitive and useful for understanding decision boundaries, it has important limitations. By construction, minimal counterfactuals tend to lie close to the model’s decision boundary. As a result, they are often *fragile*: small perturbations, noise, or model updates may revert the prediction. Moreover, such counterfactuals may correspond to atypical or low-density regions of the data distribution, leading to unrealistic or implausible recommendations. In practice, this can result in suggestions that, while technically sufficient to change the model output, do not align with how real instances of the target class appear in the data.

These limitations are particularly important when counterfactuals are used not only for explanation but also for decision support. In many applications, users are less interested in the minimal change required to flip a prediction, and more interested in changes that lead to a *stable and reliable outcome*. For example, a loan applicant may prefer guidance that leads to a clearly acceptable profile rather than one that barely crosses the approval threshold. This motivates a shift from minimal recourse to what we term *robust recourse*.

Robust recourse aims to identify counterfactuals that are not only valid, i.e., they achieve the desired prediction, but also *representative* of the target class. Intuitively, such counterfactuals lie deeper inside the decision region of the desired class and are more consistent with the empirical data distribution. This leads to explanations that are more stable, more realistic, and better aligned with human expectations of actionable recommendations.

A key challenge is that existing counterfactual generation methods, such as DiCE (Mothilal et al., 2020), typically optimize for validity, proximity, and diversity, but do not explicitly enforce distributional plausibility. As a result, they often produce multiple candidate counterfactuals without a principled way to select the most realistic one. The choice of which counterfactual to present is therefore often based on simple heuristics, such as selecting the closest or first generated instance.

In this paper, we address this limitation by introducing a selection mechanism based on conformal anomaly detection. Given a set of candidate counterfactuals, we evaluate their plausibility with respect to the empirical distribution of instances predicted as the target class. This allows us to rank candidates according to how well they align with the observed data, and to select counterfactuals that are both valid and representative. Our approach effectively transforms counterfactual explanations from minimal boundary-crossing solutions into robust, distribution-aware recourse strategies. We show empirically that, while multiple candidate counterfactuals are often available, their plausibility can vary substantially. By selecting the most plausible candidate, we obtain counterfactuals that are significantly more consistent with the data distribution, and, ideally, with only modest increases in distance or sparsity. This demonstrates that robust recourse can be achieved without sacrificing the key desirable properties of counterfactual explanations.

The contributions of this work are fourfold. First, we formalize the distinction between minimal recourse and robust recourse, arguing that minimal perturbations alone are insufficient for identifying realistic and stable counterfactual explanations. Second, we propose a conformal selection framework that evaluates candidate counterfactuals based on their conformity to the empirical target-class distribution. Third, we extend this idea beyond feature-space plausibility by introducing SHAP-space conformity, enabling robustness to be evaluated also in terms of explanatory consistency and model reasoning. Finally, we provide empirical evidence showing that substantially more conforming and robust counterfactual explanations can often be obtained with only modest increases in perturbation magnitude.

Overall, our work reframes counterfactual explanations from merely identifying boundary-crossing perturbations toward identifying representative and distributionally consistent forms of recourse. More broadly, the proposed framework illustrates how conformal methods can be used not only for uncertainty quantification, but also for evaluating the representativeness and robustness of explanations.

2. Background.

This section introduces the concepts underlying the proposed framework and positions the work within existing research on explainable artificial intelligence and uncertainty quantification. We first discuss explainability methods for machine learning models, with particular focus on SHAP explanations and counterfactual explanations. We then introduce conformal anomaly detection as the foundation for evaluating conformity and plausibility. Finally, we review related work on plausibility and robustness in counterfactual explanations, motivating the proposed conformity-based selection framework.

2.1. Explainability in Machine Learning

The increasing deployment of machine learning models in high-stakes domains has led to a growing demand for methods that provide insight into how predictions are made. This has given rise to the field of explainable artificial intelligence (XAI), which aims to make model behavior more transparent, interpretable, and trustworthy (Doshi-Velez and Kim, 2017; Molnar, 2022).

Inherently interpretable models. One approach to explainability is to design models that are interpretable by construction. Examples include linear models, decision trees, and rule-based systems. These models provide direct insight into their decision-making process, as their structure can be readily understood by humans. For instance, coefficients in linear models indicate the direction and magnitude of feature effects, while decision trees and rule lists express decisions as sequences of logical conditions. Such models are often referred to as *inherently interpretable* or *transparent* models (Rudin, 2019; Lipton, 2018).

While inherently interpretable models offer clear advantages in terms of transparency, they may be limited in their predictive performance compared to more complex models such as ensemble methods or (deep) neural networks. This trade-off has motivated the development of post-hoc explainability techniques for black-box models.

Global explainability. Global explainability methods aim to provide an overall understanding of a model’s behavior across the entire input space. These methods seek to answer questions such as which features are most important for predictions, how the model responds to changes in inputs, and what general patterns it has learned. Examples include feature importance measures, partial dependence plots, and surrogate models that approximate the behavior of a complex model using a simpler, interpretable one (Friedman, 2001; Ribeiro et al., 2016; Molnar, 2022).

Global explanations are useful for model validation, debugging, and gaining high-level insights, but they may obscure important local variations in model behavior.

Local explainability. In contrast, local explainability focuses on explaining individual predictions. Given a specific instance, local methods aim to answer why the model produced a particular output for that instance. This is particularly important in decision-making scenarios where individual predictions must be justified.

A prominent class of local explanation methods is *feature attribution*, which assigns an importance value to each feature for a given prediction. Methods such as LIME (Ribeiro et al., 2016) and SHAP (Lundberg and Lee, 2017) approximate or decompose the model locally to quantify feature contributions. These methods provide intuitive explanations in terms of how each feature influences the prediction.

Another class of local explanations is *counterfactual explanations*, which describe how an instance would need to change in order to obtain a different prediction (Wachter et al., 2018). Rather than attributing importance to existing features, counterfactuals provide actionable insights by

identifying alternative scenarios that lead to a desired outcome. In this work, we focus on counterfactual explanations, which are discussed in detail in Section 2.3.

Overall, these different approaches to explainability provide complementary perspectives on model behavior. Inherently interpretable models offer transparency by design, global methods provide high-level insights, and local methods enable instance-specific explanations, including both feature attributions and counterfactuals.

2.2. SHAP Explanations

SHAP (SHapley Additive exPlanations) is a widely used method for explaining individual predictions of machine learning models by attributing the contribution of each feature to the model output (Lundberg and Lee, 2017). SHAP is based on concepts from cooperative game theory, in particular the Shapley value, which provides a principled way to distribute a total payoff among a set of players according to their marginal contributions.

Given a model $f : \mathcal{X} \rightarrow \mathbb{R}$ and an instance $x \in \mathcal{X}$, SHAP represents the prediction as an additive decomposition

$$f(x) = \phi_0 + \sum_{j=1}^d \phi_j,$$

where ϕ_0 is a baseline value (typically the expected model output), and ϕ_j denotes the contribution of feature j to the prediction for instance x . The values ϕ_j are computed as Shapley values, which correspond to the average marginal contribution of each feature across all possible subsets of features.

SHAP values satisfy several desirable properties, including *efficiency* (the contributions sum to the prediction), *symmetry* (features with identical contributions receive equal attribution), and *consistency* (if a model changes such that a feature’s contribution increases, its SHAP value does not decrease). These properties make SHAP a theoretically well-founded approach to feature attribution.

In practice, SHAP provides a local explanation of a prediction by quantifying how each feature influences the model output relative to a baseline. Positive values ϕ_j indicate that a feature contributes toward increasing the prediction, while negative values indicate a contribution in the opposite direction. As such, SHAP explanations can be interpreted as capturing the *reasoning* of the model for a specific instance, expressed in terms of feature-level contributions.

By mapping instances to vectors of SHAP values, it is possible to analyze model behavior in an explanation space rather than the original feature space. This representation reflects how the model combines features to produce predictions, and can therefore reveal similarities and differences in the underlying decision logic across instances. In this work, we exploit this property by considering conformity not only in feature space but also in SHAP space.

2.3. Counterfactual Explanations

Counterfactual explanations provide an intuitive and actionable form of local explanation by describing how an input instance would need to change in order to obtain a different model prediction. Given an instance x with prediction $f(x)$, a counterfactual explanation identifies an alternative instance x' such that $f(x')$ yields a desired outcome, typically corresponding to a different class label. In contrast to feature attribution methods, which explain a prediction in terms of contributions of existing features, counterfactual explanations focus on *changes* to the input that would alter the decision (Wachter et al., 2018).

A common formulation of counterfactual explanations is as an optimization problem. Given a target outcome y_{target} , the goal is to find a counterfactual instance x' that satisfies

$$f(x') = y_{\text{target}},$$

while minimizing some notion of distance between x and x' . This distance typically captures the magnitude of changes applied to the input and is often combined with additional constraints reflecting feasibility or actionability. A general objective can be expressed as

$$\min_{x'} d(x, x') + \lambda \cdot \ell(f(x'), y_{\text{target}}),$$

where $d(\cdot, \cdot)$ is a distance function and ℓ measures the discrepancy between the prediction and the desired outcome.

In practice, counterfactual generation methods often incorporate multiple objectives. These include *validity*, ensuring that the prediction changes as desired; *proximity*, encouraging minimal changes to the input; *sparsity*, promoting changes in only a small number of features; and *diversity*, providing multiple distinct counterfactual explanations (Mothilal et al., 2020). Additional constraints may enforce that generated counterfactuals respect domain-specific requirements, such as immutable features or valid ranges.

Several methods have been proposed for generating counterfactual explanations, ranging from gradient-based optimization techniques to model-agnostic approaches and search-based methods. Notable examples include the optimization-based formulation by Wachter et al. (Wachter et al., 2018), diverse counterfactual generation approaches such as DiCE (Mothilal et al., 2020), and nearest-neighbor or prototype-based methods that identify similar instances with different outcomes (Van Looveren and Klaise, 2021). These approaches differ in how they balance competing objectives and in the assumptions they make about the underlying model.

The usefulness of counterfactual explanations depends on several key properties that have been widely discussed in the literature. While different methods prioritize these properties to varying degrees, they provide a useful framework for evaluating the quality of counterfactuals:

- **Validity:** The counterfactual should achieve the desired outcome, i.e., $f(x') = y_{\text{target}}$. This is the fundamental requirement for a valid counterfactual explanation.
- **Proximity:** The counterfactual should be close to the original instance according to a chosen distance metric. This ensures that the suggested changes are minimal.
- **Sparsity:** The counterfactual should modify as few features as possible. Sparse changes are typically easier to interpret and act upon.
- **Plausibility:** The counterfactual should correspond to a realistic instance, consistent with the underlying data distribution. This avoids suggesting unlikely or infeasible feature combinations.
- **Actionability:** The suggested changes should be feasible for the user to implement, taking into account domain-specific constraints such as immutable features or practical limitations.
- **Diversity:** When multiple counterfactuals are provided, they should represent distinct alternatives, offering different ways to achieve the desired outcome.
- **Stability:** Small perturbations to the input or model should not drastically change the counterfactual, ensuring robustness of the explanation.

Counterfactual explanations are particularly appealing due to their intuitive interpretation and their potential to provide actionable insights. They have been applied in a variety of domains, including finance, healthcare, and policy-making, where understanding how to achieve a desired outcome is of central importance. At the same time, the quality and usefulness of counterfactual explanations depend critically on how well they reflect realistic and feasible changes, as well as how they align with the underlying data distribution. These aspects are discussed further in the following sections.

2.3.1. DiCE: DIVERSE COUNTERFACTUAL EXPLANATIONS

DiCE (Diverse Counterfactual Explanations) [Mothilal et al. \(2020\)](#) is a model-agnostic framework for generating multiple counterfactual explanations simultaneously. In contrast to earlier approaches that primarily focused on finding a single minimally modified counterfactual, DiCE explicitly incorporates diversity among generated explanations, thereby providing multiple alternative recourse strategies.

Given an input instance $\mathbf{x}$ and a desired target outcome y_{target} , DiCE aims to generate a set of counterfactual instances

$$\{\mathbf{x}'_1, \dots, \mathbf{x}'_K\}$$

by solving a joint optimization problem of the form

$$\min_{\mathbf{x}'_1, \dots, \mathbf{x}'_K} \sum_{k=1}^K \mathcal{L}_{\text{pred}}(f(\mathbf{x}'_k), y_{\text{target}}) + \lambda_1 \mathcal{L}_{\text{prox}}(\mathbf{x}'_k, \mathbf{x}) - \lambda_2 \mathcal{L}_{\text{div}}(\{\mathbf{x}'_k\}), \quad (1)$$

where:

- $\mathcal{L}_{\text{pred}}$ encourages the generated counterfactuals to achieve the desired prediction,
- $\mathcal{L}_{\text{prox}}$ promotes proximity between the original instance and the counterfactuals,
- $\mathcal{L}_{\text{div}}$ encourages diversity among the generated counterfactuals,
- λ_1 and λ_2 control the relative importance of the different objectives.

The proximity objective is typically based on distance measures such as ℓ_1 or ℓ_2 distance, often normalized using feature-wise statistics to account for differences in feature scales. By combining prediction validity, proximity, and diversity within a unified optimization framework, DiCE can generate multiple distinct counterfactuals corresponding to different possible ways of achieving the desired outcome.

DiCE supports both continuous and categorical features and can be applied to a wide range of predictive models, including black-box classifiers. Depending on the chosen backend and optimization method, counterfactuals may be generated using gradient-based optimization, random search, or genetic algorithms.

In this work, DiCE is used as a generator of candidate counterfactual explanations. Our focus is not on evaluating the DiCE framework itself, but rather on assessing and selecting among generated counterfactuals based on their conformity to the target-class distribution. Thus, DiCE provides a diverse set of candidate explanations, while the proposed conformal framework is used to evaluate their plausibility and robustness.

2.4. Conformal Anomaly Detection

Conformal prediction provides a framework for quantifying the uncertainty of predictions with finite-sample guarantees under the assumption of exchangeability (Vovk et al., 2005; Shafer and Vovk, 2008). A central concept in this framework is that of *conformity*, which measures how well a new instance agrees with a set of previously observed examples. Conformal anomaly detection (CAD) applies this idea to assess whether a test instance is consistent with a reference distribution, without requiring knowledge of its true label.

Let $\mathcal{Z} = \{z_1, \dots, z_n\}$ denote a set of calibration instances drawn from a distribution of interest. A nonconformity function

$$A : \mathcal{X} \rightarrow \mathbb{R}$$

assigns a score to each instance, where larger values indicate lower conformity with respect to the calibration set. The choice of nonconformity function is flexible and can be tailored to the application, for example by measuring distances to calibration instances or deviations from learned representations. In this work, we will define specific nonconformity functions in the method section.

Given the calibration set, nonconformity scores are computed as

$$\alpha_i = A(z_i), \quad i = 1, \dots, n.$$

For a new test instance x , the corresponding nonconformity score $\alpha^* = A(x)$ is evaluated relative to the empirical distribution of calibration scores. This yields a conformal p-value defined as

$$p(x) = \frac{|\{i : \alpha_i \geq \alpha^*\}| + 1}{n + 1}.$$

Intuitively, the p-value quantifies how extreme the test instance is compared to the calibration set: large values indicate that the instance is typical, while small values suggest that it deviates from the calibration distribution.

A key property of conformal p-values is their validity under exchangeability. If the test instance is drawn from the same distribution as the calibration set, then the p-values are uniformly distributed on $[0, 1]$, implying that

$$\mathbb{P}(p(x) \leq \epsilon) \leq \epsilon$$

for any $\epsilon \in (0, 1)$. This provides a principled way to assess conformity without relying on explicit probabilistic modeling assumptions.

While CAD is often used for anomaly detection by thresholding p-values, its interpretation is more general. Rather than providing a binary decision, conformal p-values offer a continuous measure of how well a test instance aligns with the empirical structure of the calibration data. In particular, CAD can be viewed as a distribution-free method for estimating the relative typicality of new instances with respect to a reference set.

An important feature of this framework is that it does not require access to the true label of the test instance. Instead, conformity is evaluated solely based on the relationship between the instance and the calibration set. This makes CAD particularly suitable in settings where the goal is to assess whether an instance resembles a certain subset of data, such as instances associated with a given model prediction or decision region.

In this work, we leverage conformal anomaly detection as a tool for evaluating how well candidate instances conform to a reference distribution. By computing p-values with respect to calibration data, we obtain a principled, distribution-free measure of conformity that can be used to compare and rank instances according to their consistency with observed data.

2.5. Related Work: Plausibility in Counterfactual Explanations

A central challenge in counterfactual explanations is ensuring that generated instances correspond to realistic and meaningful alternatives. While early formulations primarily focused on validity and proximity (Wachter et al., 2018), it has become increasingly clear that these criteria alone are insufficient to guarantee useful explanations. In particular, counterfactuals that lie in low-density or implausible regions of the feature space may be difficult to interpret or act upon, even if they achieve the desired prediction.

Several approaches have been proposed to encourage plausibility in counterfactual generation. One line of work incorporates explicit constraints derived from domain knowledge, such as restricting changes to actionable features or enforcing feasibility conditions (Ustun et al., 2019; Karimi et al., 2021). These approaches ensure that counterfactuals respect practical or causal constraints, but they do not directly address whether generated instances resemble typical observations from the data distribution.

Another direction focuses on modeling the data manifold. Methods based on generative models, such as variational autoencoders or generative adversarial networks, aim to produce counterfactuals that remain close to the learned data distribution (Dhurandhar et al., 2018; Joshi et al., 2019). Similarly, prototype-based approaches generate counterfactuals by referencing existing instances or learned prototypes (Van Looveren and Klaise, 2021). While these methods improve plausibility by construction, they often require additional modeling assumptions and may be sensitive to the quality of the learned generative model.

More recently, several works have emphasized the importance of distributional and causal consistency. For example, causal approaches aim to generate counterfactuals that respect underlying causal relationships between features (Karimi et al., 2020; Ma et al., 2022). Other methods incorporate density estimation or likelihood-based criteria to penalize counterfactuals that lie in low-density regions (Poyiadzi et al., 2020; Artelt et al., 2021). These approaches move toward capturing the notion of plausibility as consistency with the data-generating process, but often rely on explicit density models or graph structures.

A particularly relevant recent direction concerns evaluating counterfactual explanations through their coherence with real instances in explanation space. In (Johansson and Maalej, 2026), plausibility was defined not only in terms of feature-space similarity, but also in terms of *explanatory coherence*. Specifically, the work introduced SHAP-space conformity as a mechanism for evaluating whether the explanatory structure of a counterfactual resembles that of genuine instances from the target class. Using conformal anomaly detection with several nonconformity measures, the study demonstrated that real instances exhibit approximately calibrated rejection rates, while generated counterfactuals are rejected substantially more often, indicating limited coherence in SHAP space.

The purpose of this paper is different from (Johansson and Maalej, 2026) in an important way. Rather than using conformal methods primarily to evaluate the plausibility of generated counterfactuals, we use conformity scores actively as a selection mechanism for identifying more representative and less fragile forms of recourse among multiple candidate counterfactuals.

We have also previously combined counterfactual explanations with conformal prediction frameworks. In (Maalej et al., 2025), counterfactual explanations were generated for conformal prediction sets in classification, while (Maalej and Johansson, 2026) extended this idea to conformal regression intervals. In these works, the objective was to explain the uncertainty-aware outputs produced by conformal predictors, i.e., prediction sets and prediction intervals, rather than individual point predictions from standard classifiers. Consequently, the generated counterfactuals aimed to alter the conformal output itself, for example by changing the size or composition of a prediction set.

Here, the present work differs substantially in both objective and methodology. Rather than generating counterfactual explanations for conformal predictions, we use conformal methods to evaluate and select among candidate counterfactual explanations generated for ordinary predictive models. Thus, conformal prediction is here used as a mechanism for assessing distributional conformity and robustness of candidate counterfactuals, rather than as the object being explained.

Despite these advances, existing methods typically integrate plausibility into the generation process itself, either through constraints, regularization, or generative modeling. In contrast, comparatively little attention has been given to the problem of *selecting* among multiple candidate counterfactuals based on their consistency with observed data. In practice, many counterfactual generation methods produce a set of valid candidates that satisfy proximity and sparsity constraints, but do not provide a principled mechanism for identifying the most realistic alternative.

Our work adopts a complementary perspective. Rather than modifying the generation process, we focus on evaluating and selecting counterfactuals based on their conformity to an empirical reference distribution. By leveraging conformal anomaly detection, we obtain a distribution-free measure of how well candidate counterfactuals align with observed instances associated with the target outcome. This enables a principled comparison of counterfactuals in terms of their typicality, without requiring explicit density estimation or generative modeling assumptions.

This perspective connects counterfactual explanations with broader themes in uncertainty quantification and distributional conformity. In particular, it allows counterfactuals to be interpreted not only as minimal changes that alter a prediction, but also as instances that are representative of the target class, thereby supporting more robust and realistic recourse.

3. Method.

This section presents the proposed framework for conformity-based counterfactual selection. We first describe how candidate counterfactuals are evaluated and selected using conformal anomaly detection in both feature space and SHAP space. We then describe the experimental setup, datasets, and evaluation protocol used in the empirical study.

3.1. Conformal Selection of Counterfactual Explanations

The main idea of this work is to move from minimal recourse toward recourse that is less fragile and more representative of the target-class distribution. This is achieved by selecting counterfactual explanations that are not only valid, but also highly conforming to the empirical target-class distribution. In the proposed framework, robustness is operationalized as conformity to the empirical target-class distribution. Consequently, a high-quality counterfactual is not merely one that flips the prediction, but one that also resembles typical target-class instances according to the predictive model. So, rather than presenting the closest generated counterfactual, we instead select the counterfactual that achieves the highest conformity according to conformal anomaly detection.

Let $\mathbf{x}$ denote an original instance with predicted label

$$\hat{y} = f(\mathbf{x}),$$

and let

$$\mathcal{C}(\mathbf{x}) = \{\mathbf{x}'_1, \dots, \mathbf{x}'_K\}$$

denote a set of candidate counterfactual explanations generated toward the opposite predicted class,

$$y_{\text{target}} = 1 - \hat{y},$$

in the binary classification setting. Candidate counterfactuals that do not achieve the desired prediction flip are discarded, such that all retained candidates satisfy

$$f(\mathbf{x}'_k) = y_{\text{target}}.$$

Traditional counterfactual methods typically select the counterfactual with smallest distance to the original instance,

$$\arg \min_{\mathbf{x}'_k \in \mathcal{C}(\mathbf{x})} d(\mathbf{x}, \mathbf{x}'_k),$$

thereby emphasizing minimal perturbation. In contrast, the proposed approach selects the counterfactual with highest conformity to the target-class distribution,

$$\arg \max_{\mathbf{x}'_k \in \mathcal{C}(\mathbf{x})} p(\mathbf{x}'_k),$$

where $p(\cdot)$ is a conformal p-value computed using CAD.

The underlying intuition is that counterfactuals located close to the decision boundary may correspond to fragile or atypical instances of the target class. By instead selecting highly conforming counterfactuals, the resulting explanations become more representative of the target-class region and therefore provide more robust recourse.

In the proposed framework, conformity is evaluated separately in feature space and SHAP space. Feature-space conformity measures whether a counterfactual resembles typical target-class instances in terms of feature values, while SHAP-space conformity measures whether the explanatory pattern underlying the prediction resembles the predictive logic commonly associated with the target class. Consequently, the proposed approach allows robust recourse to be defined both in terms of data-space plausibility and explanatory consistency.

Calibration sets. To evaluate conformity with respect to the target class, calibration sets are constructed separately for each predicted class. Let

$$\mathcal{Q}_c = \{\mathbf{q}_1, \dots, \mathbf{q}_{n_c}\}$$

denote the set of calibration instances predicted as class c , i.e.,

$$f(\mathbf{q}_i) = c.$$

These calibration sets define the empirical conformity distributions used for computing conformal p-values. Since conformity is evaluated conditionally on the predicted class, a counterfactual assigned to class c is always compared only to calibration instances predicted as class c . This choice is intentional, since the goal is to assess conformity relative to the model-defined target region that the counterfactual aims to enter, rather than relative to the true class labels.

Feature-space conformity. In feature space, conformity is evaluated based on distances to nearby target-class instances. Let

$$d(\mathbf{x}, \mathbf{q})$$

denote Euclidean distance in standardized feature space. For an instance $\mathbf{x}$, the k nearest neighbors among the calibration instances of the same predicted class are identified, and their mean vector is computed as

$$\boldsymbol{\mu}_k(\mathbf{x}) = \frac{1}{k} \sum_{i=1}^k \mathbf{q}_{(i)},$$

where

$$\mathbf{q}_{(i)}$$

denotes the i -th nearest calibration instance. The feature-space nonconformity score is then defined as

$$\alpha_{\text{feat}}(\mathbf{x}) = \|\mathbf{x} - \boldsymbol{\mu}_k(\mathbf{x})\|_2.$$

This nonconformity measure captures how well an instance aligns with the local feature-space structure of the target class. Small values correspond to instances located near the center of nearby target-class instances, while large values indicate atypical or isolated points.

SHAP-space conformity. While feature-space conformity evaluates similarity in terms of raw input features, SHAP-space conformity evaluates similarity in terms of the model’s reasoning. Let

$$\boldsymbol{\phi}(\mathbf{x}) = (\phi_1(\mathbf{x}), \dots, \phi_d(\mathbf{x}))$$

denote the SHAP explanation vector for instance $\mathbf{x}$. The SHAP-space nonconformity score is defined analogously. First, the mean SHAP vector of the k nearest calibration explanations is computed as

$$\boldsymbol{\mu}_k^\phi(\mathbf{x}) = \frac{1}{k} \sum_{i=1}^k \boldsymbol{\phi}(\mathbf{q}_{(i)}).$$

The SHAP-space nonconformity score is then given by

$$\alpha_{\text{SHAP}}(\mathbf{x}) = \left\| \boldsymbol{\phi}(\mathbf{x}) - \boldsymbol{\mu}_k^\phi(\mathbf{x}) \right\|_2.$$

Thus, SHAP-space conformity measures whether the model explains an instance in a manner similar to how it explains nearby target-class instances. This provides a notion of explanatory conformity rather than merely geometric similarity in feature space.

Conformal p-values. For both feature-space and SHAP-space conformity, calibration nonconformity scores are computed for all calibration instances of a given predicted class,

$$\alpha_1, \dots, \alpha_n.$$

Given a candidate counterfactual with nonconformity score α^* , the conformal p-value is computed as

$$p(\mathbf{x}) = \frac{|\{i : \alpha_i > \alpha^*\}| + 1}{n + 1}.$$

Large p-values indicate that the counterfactual conforms well to the target-class distribution, while small p-values indicate atypical or weakly conforming instances.

Finally, among the generated candidate counterfactuals, we select the instance with highest conformity according to either feature-space or SHAP-space CAD. This transforms counterfactual explanations from minimally sufficient boundary-crossing instances into more robust and distributionally consistent recourse strategies.

3.2. Experimental Setup and Evaluation Protocol

Three experiments were conducted to evaluate the proposed framework.

Experiment 1: CAD sanity check. The first experiment evaluated whether the proposed CAD framework exhibits the expected conformal behavior on naturally occurring test instances. Specifically, we investigated whether rejection rates approximately matched the target significance levels ϵ . This experiment serves as a validation of the proposed conformity measures in both feature space and SHAP space. For each dataset, a 10-fold cross-validation protocol was used. Within each fold, the training portion was further divided into model-training and calibration sets. CAD was then evaluated on all test instances, conditioning on the predicted class label. Rejection rates were computed for $\epsilon \in \{0.05, 0.10, 0.20\}$ and results were pooled across folds.

Experiment 2: Illustrative instance-level examples. The second experiment provides qualitative illustrations of the proposed framework at the individual-instance level. For selected factual instances, multiple candidate counterfactuals were generated and evaluated using both feature-space and SHAP-space CAD. The experiment illustrates how counterfactuals with similar proximity may differ substantially in terms of conformity and robustness. In this experiment, counterfactuals were generated using the random-search variant of DiCE.

Experiment 3: Counterfactual selection and robust recourse. The main experiment evaluated the proposed conformal selection framework by comparing the selected counterfactual to the closest generated counterfactual. The purpose was to investigate whether substantially more conforming counterfactuals can be obtained with only modest increases in distance from the original instance. For each factual instance, DiCE generated a set of candidate counterfactuals targeting the opposite predicted class. Among valid counterfactuals, two selection strategies were compared:

- **Closest counterfactual:** the candidate with smallest normalized ℓ_1 -distance to the original instance,
- **Conformal selection:** the candidate with highest conformal p-value.

The evaluation focused on the trade-off between proximity and robustness. In particular, we report:

- conformity p-values,
- rejection rates,
- normalized ℓ_1 -distances from the factual instance,
- relative increase in distance compared to the closest counterfactual.

Experiment 3 used the same 10-fold cross-validation protocol as Experiment 1. To reduce computational cost, counterfactual generation and evaluation were performed only for the final fold of each dataset.

Datasets. Experiments 1 and 3 used twelve regression-style benchmark datasets: the four kin8, the four bank8, and the four puma8 datasets. Each dataset contains eight continuous features and 8192 instances. To enable binary classification, target values above the dataset mean were assigned to class 1, while remaining instances were assigned to class 0. Experiment 2 used the Pima Indians Diabetes dataset, which is commonly used in counterfactual explanation research.

Predictive models and explanations. All experiments used Random Forest classifiers with 300 trees. For CAD, the predicted class label was used to determine the calibration category. Consequently, conformity was evaluated relative to the model-defined target-class region rather than the true class labels. SHAP explanations were computed using TreeSHAP, producing feature-level contribution vectors for each instance. These vectors were used to construct SHAP-space conformity measures.

Counterfactual generation. Counterfactuals were generated using the DiCE framework. Unless otherwise stated, default settings were used, and 10 candidate counterfactuals were generated for each factual instance using the opposite predicted class as the target outcome. Experiment 2 used the random-search backend of DiCE, while Experiment 3 used the genetic-search backend.

To improve realism and avoid implausible feature values, generated counterfactuals were constrained to remain within feature-wise ranges observed in the training data. In addition, only selected actionable features were allowed to vary in Experiment 2.

4. Results.

This section presents the empirical evaluation of the proposed framework. We first validate the conformal behavior of the proposed conformity measures through a CAD sanity check on real instances. We then provide qualitative instance-level examples illustrating conformity-based counterfactual selection in both feature space and SHAP space. Finally, we evaluate the proposed selection framework quantitatively by comparing the selected counterfactuals to the closest generated counterfactuals across multiple benchmark datasets.

4.1. Experiment 1: The CAD-procedure

We start by investigating the CAD-procedure. Table 1 presents the results of the CAD sanity-check experiment for both feature-space and SHAP-space conformity measures.

Table 1: Experiment 1: Empirical rejection rates for conformal anomaly detection in feature space and SHAP space across all datasets. Results are shown for significance levels $\epsilon \in \{0.05, 0.10, 0.20\}$. The empirical rejection rates closely match the corresponding target significance levels, indicating that the conformal procedure is well calibrated in both spaces. Mean values across all datasets are reported in the final row group.

Dataset	Accuracy	$\epsilon = 0.05$		$\epsilon = 0.10$		$\epsilon = 0.20$	
		Feat.	SHAP	Feat.	SHAP	Feat.	SHAP
bank8nm	0.912	0.046	0.048	0.093	0.097	0.191	0.198
bank8nh	0.783	0.048	0.045	0.098	0.099	0.196	0.201
bank8fm	0.938	0.046	0.049	0.096	0.101	0.198	0.204
bank8fh	0.844	0.049	0.044	0.099	0.093	0.198	0.195
puma8nm	0.935	0.049	0.049	0.099	0.096	0.208	0.199
puma8nh	0.826	0.051	0.052	0.099	0.101	0.199	0.193
puma8fm	0.929	0.049	0.049	0.101	0.101	0.201	0.197
puma8fh	0.798	0.045	0.049	0.095	0.100	0.199	0.202
kin8nm	0.829	0.047	0.050	0.102	0.104	0.197	0.200
kin8nh	0.772	0.050	0.047	0.102	0.095	0.201	0.196
kin8fm	0.916	0.053	0.047	0.099	0.099	0.206	0.193
kin8fh	0.819	0.052	0.047	0.102	0.097	0.197	0.194
Mean	0.858	0.049	0.048	0.099	0.099	0.199	0.198

Overall, the empirical rejection rates closely match the target significance levels $\epsilon \in \{0.05, 0.10, 0.20\}$, indicating that the proposed CAD framework behaves as intended. This behavior is observed consistently across all twelve datasets and for both conformity spaces. Averaged across datasets, the empirical rejection rates were 0.051, 0.098, and 0.201 for feature-space CAD, compared to target values of 0.05, 0.10, and 0.20, respectively. Corresponding values for SHAP-space CAD were 0.050,

0.100, and 0.200. These results indicate that the conformal p-values are well calibrated and that the proposed conformity measures provide approximately valid assessments of target-class conformity.

The results also demonstrate that CAD behaves similarly in both feature space and SHAP space, despite SHAP-space conformity being based on explanation vectors rather than the original feature representation. This supports the interpretation of SHAP-space CAD as a meaningful measure of explanatory conformity.

4.2. Experiment 2: Illustrative Examples

To complement the quantitative evaluation, Experiment 2 provides qualitative illustrations of the proposed framework at the individual-instance level. The purpose of this experiment is to visualize how different counterfactual selection strategies may lead to substantially different forms of recourse, despite all generated counterfactuals satisfying the desired target prediction.

The experiment used the Pima Indians Diabetes dataset, a widely used benchmark dataset in research on explainable artificial intelligence and counterfactual explanations.

Here, only actionable features were allowed to vary during counterfactual generation, specifically glucose level, insulin level, and body mass index (BMI). This restriction was introduced to improve realism and avoid implausible recourse strategies involving immutable or clinically less actionable variables. Figure 1 presents a representative example.

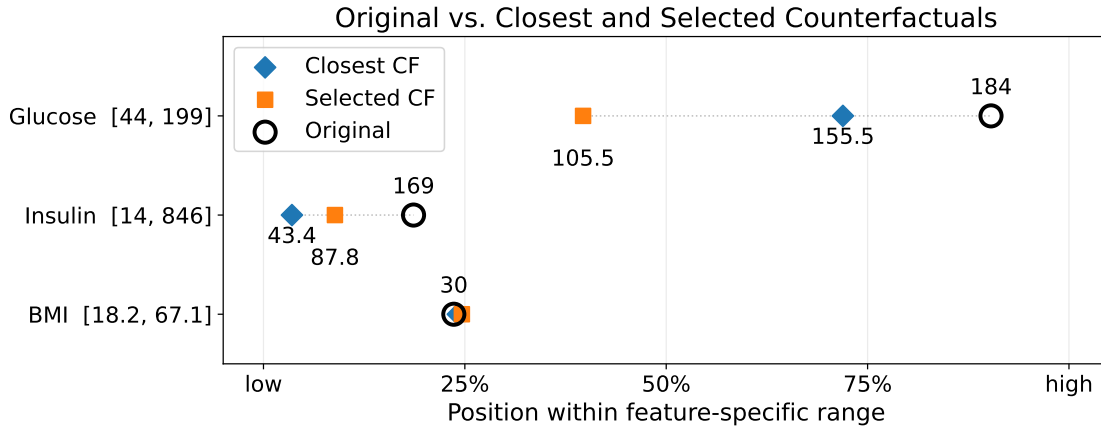


Figure 1: Example of conformal counterfactual selection in feature space for the Pima Indians Diabetes dataset. The factual instance is shown together with the closest generated counterfactual and the selected counterfactual with highest conformal p-value. Although both counterfactuals achieve the desired target prediction, the selected counterfactual exhibits substantially higher conformity and classifier confidence while requiring only moderately larger perturbations.

The factual instance was predicted as diabetic with confidence 0.767. The closest generated counterfactual achieved the desired target prediction but obtained a relatively low conformal p-value of 0.036, causing it to be rejected at significance level $\epsilon = 0.05$. In addition, the classifier confidence, i.e., the predicted probability for the target class, remained relatively weak (0.537), suggesting that the counterfactual lies close to the decision boundary.

In contrast, the selected counterfactual achieved a substantially higher conformal p-value of 0.132 while increasing the normalized ℓ_1 -distance by only a factor of approximately 1.5. The classifier confidence also increased substantially to 0.777, indicating that the selected counterfactual lies deeper within the target-class region.

This example illustrates an important conceptual difference between minimal and robust recourse. The closest counterfactual primarily performs the smallest possible perturbation necessary to cross the decision boundary, whereas the selected counterfactual produces a more coherent target-class profile by simultaneously lowering glucose, insulin, and BMI. Thus, the selected counterfactual better aligns with the empirical structure of the target class rather than merely satisfying the prediction constraint.

A second example illustrating SHAP-space conformity is shown in Figure 2. In this case, the factual instance was predicted as non-diabetic with high confidence (0.877). The closest generated counterfactual achieved the desired target prediction, but obtained a very low SHAP-space conformal p-value of only 0.011, indicating that the model’s reasoning for this instance differed substantially from the reasoning typically observed for genuine target-class instances. In addition, the classifier confidence for the target class remained close to the decision boundary (0.507).

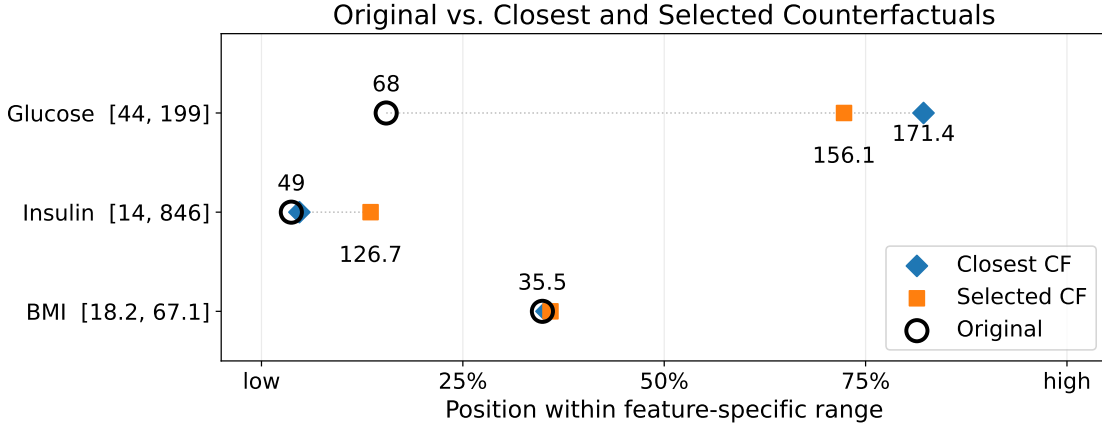


Figure 2: Example of conformal counterfactual selection in SHAP space for the Pima Indians Diabetes dataset. Although the closest and selected counterfactuals exhibit similar feature-space proximity, they differ substantially in explanatory conformity. The selected counterfactual achieves dramatically higher SHAP-space conformity and classifier confidence while requiring only marginally larger perturbations.

In contrast, the selected counterfactual achieved an extremely high SHAP-space p-value of 0.978, while increasing the normalized ℓ_1 -distance by only approximately 10%. The classifier confidence also increased substantially to 0.743. Importantly, the geometric differences between the closest and selected counterfactuals are relatively modest, yet the explanatory conformity differs dramatically.

This example illustrates a key motivation for SHAP-space CAD. Two counterfactuals may exhibit very similar feature-space proximity while corresponding to fundamentally different forms of model reasoning. The closest counterfactual represents a fragile boundary-crossing instance with atypical explanatory structure, whereas the selected counterfactual aligns much more closely with the explanatory patterns associated with typical target-class instances. Thus, SHAP-space selection provides a notion of robustness that cannot be captured by proximity alone.

Together, these two examples show how conformal selection can transform counterfactual explanations from fragile boundary-crossing instances into more representative and robust forms of recourse.

4.3. Experiment 3: Large-scale Evaluation

Table 2 presents the results of the proposed conformal counterfactual selection framework. Across all datasets, selecting the counterfactual with highest conformity substantially increased the mean conformal p-values compared to selecting the closest counterfactual. This behavior was observed consistently for both feature-space and SHAP-space CAD.

Table 2: Experiment 3: Comparison between the closest counterfactual explanation and the conformally selected counterfactual explanation using feature-space and SHAP-space CAD. For each space, we report the mean conformal p-value for the closest counterfactual (p_{close}) and the selected counterfactual (p_{sel}), rejection rates at $\epsilon = 0.10$, the mean normalized ℓ_1 distance of the closest counterfactual to the factual instance ($\text{Dist}_{\text{close}}$), and the relative distance of the selected counterfactual compared to the closest counterfactual ($\text{RelDist}_{\text{sel}}$).

Dataset	Feature-space CAD						SHAP-space CAD					
	p_{close}	p_{sel}	$\text{Rej}_{\text{close}}$	Rej_{sel}	$\text{Dist}_{\text{close}}$	$\text{RelDist}_{\text{sel}}$	p_{close}	p_{sel}	$\text{Rej}_{\text{close}}$	Rej_{sel}	$\text{Dist}_{\text{close}}$	$\text{RelDist}_{\text{sel}}$
bank8nm	0.467	0.891	0.115	0.000	3.313	1.426	0.198	0.600	0.495	0.007	3.313	1.449
bank8nh	0.287	0.787	0.322	0.004	2.764	1.539	0.117	0.513	0.694	0.054	2.764	1.543
bank8fm	0.457	0.843	0.155	0.004	3.404	1.407	0.109	0.416	0.651	0.027	3.404	1.516
bank8fh	0.294	0.727	0.338	0.007	2.948	1.560	0.056	0.289	0.844	0.299	2.948	1.622
puma8nm	0.362	0.842	0.284	0.011	3.126	1.384	0.220	0.754	0.503	0.009	3.126	1.413
puma8nh	0.312	0.831	0.370	0.009	2.995	1.427	0.135	0.670	0.683	0.037	2.995	1.439
puma8fm	0.420	0.878	0.187	0.000	3.329	1.341	0.184	0.724	0.530	0.001	3.329	1.366
puma8fh	0.259	0.830	0.424	0.000	3.060	1.424	0.072	0.518	0.807	0.050	3.060	1.453
kin8nm	0.350	0.811	0.300	0.002	3.070	1.303	0.210	0.640	0.484	0.009	3.070	1.291
kin8nh	0.269	0.774	0.418	0.005	2.963	1.308	0.184	0.601	0.584	0.035	2.963	1.299
kin8fm	0.406	0.861	0.181	0.001	3.385	1.260	0.282	0.708	0.299	0.000	3.385	1.264
kin8fh	0.299	0.791	0.350	0.001	3.089	1.319	0.265	0.670	0.349	0.000	3.089	1.291
Mean	0.349	0.822	0.287	0.004	3.121	1.392	0.169	0.592	0.577	0.044	3.121	1.412

For feature-space CAD, the mean p-value increased from 0.349 for the closest counterfactuals to 0.822 for the selected counterfactuals. At the same time, the rejection rate decreased dramatically from 28.7% to only 0.4%. Similarly, in SHAP space, the mean p-value increased from 0.169 to 0.592, while the rejection rate decreased from 57.7% to 4.4%. Thus, the proposed selection strategy consistently identified substantially more conforming and robust counterfactual explanations than simply selecting the closest candidate.

These improvements in conformity and distributional robustness were obtained at the cost of moderately increased distance from the original instance. On average, the selected counterfactuals were approximately 1.39 times farther away than the closest counterfactuals in feature-space selection and 1.41 times farther away in SHAP-space selection. Hence, relatively limited increases in perturbation magnitude yielded very large improvements in conformity and robustness.

The results also reveal an important distinction between feature-space and SHAP-space conformity. While the closest counterfactuals often exhibited moderate conformity in feature space, their SHAP-space conformity was substantially lower, with rejection rates frequently exceeding 50%. This indicates that many minimally perturbed counterfactuals, despite achieving the desired prediction flip, are associated with atypical explanatory patterns relative to genuine instances of the target class. In other words, the model may arrive at the target prediction through reasoning patterns that are uncommon for that class.

The proposed SHAP-space selection mechanism mitigates this issue by explicitly favoring counterfactuals whose explanatory structure resembles that of typical target-class instances. Consequently, the selected counterfactuals not only achieve the desired prediction outcome, but also align more closely with the local predictive logic normally associated with the target class. This supports the interpretation of the selected explanations as more robust forms of recourse, moving beyond purely minimal perturbations toward counterfactuals that better reflect the broader target-class manifold.

5. Concluding Remarks.

This paper proposed a conformal framework for selecting counterfactual explanations based not only on proximity to the original instance, but also on conformity to the empirical target-class distribution. The main motivation was to move beyond the traditional notion of *minimal recourse* toward more *robust* and representative forms of recourse. Rather than selecting the closest counterfactual among generated candidates, the proposed framework selects the counterfactual with highest conformal p-value according to conformal anomaly detection (CAD).

The empirical results demonstrated that substantially more conforming counterfactual explanations can often be obtained with only modest increases in perturbation magnitude. Across all evaluated benchmark datasets, the selected counterfactuals consistently achieved much higher conformity than the closest counterfactuals, both in feature space and in SHAP space. In particular, SHAP-space CAD revealed that counterfactuals with similar geometric proximity may nevertheless differ substantially in terms of explanatory conformity. This suggests that minimally perturbed counterfactuals may rely on atypical or weakly conforming model reasoning patterns, despite satisfying the desired prediction outcome. The qualitative examples further illustrated how conformity-based selection can produce explanations that appear more representative of genuine target-class instances. More generally, the results suggest that conformal methods may provide a useful foundation not only for uncertainty quantification, but also for evaluating the representativeness and explanatory consistency of AI explanations and recourse strategies.

The proposed framework has several limitations. First, conformity is evaluated relative to the predictive model itself, implying that the quality of the selected counterfactuals depends on the quality and calibration of the underlying model. Second, SHAP-space conformity inherits limitations associated with SHAP explanations and therefore reflects the explanatory behavior of the model rather than causal mechanisms in the data-generating process. Third, the proposed conformity measures are based on local nearest-neighbor structure and may therefore depend on calibration-set density and feature representation. Finally, while conformity-based selection often required only moderate increases in distance, there remains an inherent trade-off between minimality and robustness. Highly conforming counterfactuals may in some cases require substantially larger or less actionable interventions.

Several directions for future work remain. An important extension is to investigate alternative conformity measures and distance metrics, including causal or domain-informed notions of plausibility. Another promising direction is to combine conformity-based selection with explicit actionability constraints and user preferences. In addition, while this work focused on static predictive settings, future studies could investigate robustness under model updates or distribution shift. Finally, the proposed framework is independent of the underlying counterfactual generator and could therefore be integrated with a wide range of existing recourse methods and explanation frameworks.

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